

Computer Vision Assignment 1: Image Stitching Pipeline

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1 Objective

The objective of this assignment was to create a manual image stitching pipeline to combine multiple (at least three) overlapping images into a single, unified panoramic image. The task required implementing feature detection, geometric transformation (Homography), and image warping using standard libraries (OpenCV, NumPy) without relying on high-level automated stitching APIs.

2 Methodology

The project was implemented using Python. The pipeline followed these core steps:

2.1 1. Data Collection

Three images of our hostel were captured using a smartphone camera.

- The camera was rotated around a fixed optical center to minimize parallax errors.
- An overlap of approximately 30-40% was maintained between consecutive images to ensure sufficient shared features for matching.

2.2 2. Feature Extraction (SIFT)

The Scale-Invariant Feature Transform (SIFT) algorithm was utilized to detect keypoints and compute descriptors. SIFT was chosen for its invariance to scale and rotation, which is crucial when stitching images taken from different angles.

- **Keypoints:** Distinctive locations in the image (corners, textured areas).
- **Descriptors:** Vectors describing the visual appearance of the patch around a keypoint.

2.3 3. Feature Matching Filtering

Features between image pairs (Left-Center and Right-Center) were matched using a Brute-Force Matcher.

- **Lowe’s Ratio Test:** To ensure robustness, matches were filtered using Lowe’s Ratio Test (threshold = 0.75). This discarded ambiguous matches where the distance to the nearest neighbor was not significantly better than the second-nearest neighbor.

2.4 4. Homography Estimation (RANSAC)

Using the filtered good matches, a Homography matrix (H) was computed to map pixels from the source image to the destination image.

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (1)$$

RANSAC (Random Sample Consensus) was applied within the `cv2.findHomography` function. This was critical for ignoring ”outlier” matches (noise) and ensuring the transformation matrix was calculated based only on geometrically consistent points.

2.5 5. Warping and Blending

The images were warped into a common coordinate system using the computed homography matrices.

- A translation matrix was applied to ensure that pixels warped to negative coordinates were shifted into the visible canvas area.
- A blending strategy using `np.maximum` was applied to merge the overlapping regions, prioritizing the most prominent pixel values to reduce ghosting artifacts.

3 Results and Comparison

3.1 Naive Stitch (Direct Overlay)

The figure below demonstrates the result of a ”naive” stitch, where images were simply warped and overlaid without blending logic.



Figure 1: Naive Stitching Result. Notice the distinct vertical seams (visible as intensity jumps) where the images overlap. The transition is abrupt and visually jarring.

3.2 Final Processed Panorama

The figure below shows the final output after applying the full pipeline with robust warping and blending.



Figure 2: Final Processed Panorama. The sharp seams have been significantly mitigated. The structural lines of the building (windows, balconies) are aligned correctly, demonstrating accurate homography estimation.

4 Discussion

- **Geometric Alignment:** As seen in the Final Panorama (Fig. 2), the horizontal lines of the building floors remain straight across the stitch boundaries. This confirms that the RANSAC algorithm successfully found the correct transformation despite perspective distortion.
- **Seam Removal:** The naive stitch (Fig. 1) clearly shows where one image ends and the next begins due to slight exposure differences in the sky and walls. The final processing step smoothed these transitions, creating a more cohesive image.
- **Challenges:** The primary challenge was ensuring the warped images did not get cut off. This was solved by calculating the bounding box of the final canvas and applying a translation offset before warping.

5 Conclusion

The assignment successfully demonstrated the fundamental principles of computer vision stitching. By manually implementing feature matching and homography estimation, a seamless panoramic view was constructed from individual photographs.